

CodeAlpha Internship Project Report

Domain:	Artificial Intelligence	Academic Year:	B.Tech 2nd Year
Project Title:	Task 2 - Chatbot for FAQs	Submission Date:	June 2026

1. Project Overview

Yeh ek NLP-based FAQ Chatbot hai jo user ke puche gaye sawaal (queries) ko samajhta hai aur database mein saved predefined FAQs se match karke sabse best aur accurate answer return karta hai. Is project mein Basic Natural Language Processing (NLP) techniques ka use kiya gaya hai taaki similarity match ho sake.

2. Technical Approach & Logic

Ek B.Tech 2nd year student ke level ke hisab se isme scikit-learn ki TF-IDF (Term Frequency-Inverse Document Frequency) Vectorization technique aur **Cosine Similarity** ka use kiya gaya hai:

- **Dataset:** Ek dictionary banayi gayi hai jisme college ya general system se related FAQs add kiye hain.
- **Text Processing:** User ke question ko TF-IDF vectors me transform kiya jata hai aur check kiya jata hai ki database ke kis question se iski mathematical similarity sabse zyada hai.
- **Fallback Mechanism:** Agar similarity score ek specific threshold (e.g., 0.2) se kam aata hai, toh chatbot "I'm sorry, I didn't understand that." display karta hai.

3. Python Source Code

```
import numpy as np
from sklearn.feature_extraction.text import TfidfVectorizer
from sklearn.metrics.pairwise import cosine_similarity

# Predefined FAQ Dataset (Questions and Answers)
faq_data = {
    "what is your name?": "I am CodeAlpha Support Chatbot.",
    "how can I submit my internship project?": "You can submit your project via the submission form shared in your official WhatsApp group.",
    "what are the passing criteria for this internship?": "You must complete at least 2 or 3 tasks from the given task list to become eligible for the completion certificate.",
    "when will I get my completion certificate?": "Certificates are usually processed and shared via email or your group within 7-14 days after successful verification of your tasks.",
    "can I get a letter of recommendation?": "Yes, high-performing candidates receive a performance-based Letter of Recommendation (LOR).",
    "how to contact support?": "You can contact support via email at
```

```

services.codealpha@gmail.com or WhatsApp."
}

questions = list(faq_data.keys())
answers = list(faq_data.values())

# Initialize TF-IDF Vectorizer
vectorizer = TfidfVectorizer(stop_words='english')
tfidf_matrix = vectorizer.fit_transform(questions)

def get_chatbot_response(user_query):
    # Vectorize user input
    user_vector = vectorizer.transform([user_query])

    # Calculate cosine similarity against all dataset questions
    similarities = cosine_similarity(user_vector, tfidf_matrix).flatten()

    # Find the index of highest similarity
    best_match_idx = np.argmax(similarities)
    max_similarity = similarities[best_match_idx]

    # Define threshold for relevance
    if max_similarity > 0.25:
        return answers[best_match_idx]
    else:
        return "I am sorry, I couldn't find an exact answer to that in our FAQs.
Please email support at services.codealpha@gmail.com."

# Console Interface for testing
def main():
    print("="*50)
    print("      Welcome to CodeAlpha FAQ Chatbot (AI Task 2)      ")
    print("      (Type 'quit' or 'exit' to stop the chat)      ")
    print("="*50)

    while True:
        user_input = input("
You: ").strip()
        if user_input.lower() in ['quit', 'exit']:
            print("Chatbot: Goodbye! Have a great day.")
            break
        if user_input == "":
            continue

        response = get_chatbot_response(user_input)
        print(f"Chatbot: {response}")

if __name__ == "__main__":
    main()

```

4. How to Run

Install the required standard data science and NLP tools using pip:

- `pip install scikit-learn numpy`